AA-C-553d
September 14, 1965

SUPERSEDING
Fed. Spec. AA-C-553c
September 23, 1959

FEDERAL SPECIFICATION

COSTUMER, WEARING APPAREL, METAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 The metal costumers covered by this specification shall be of one type and one size as shown in figure 1.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-490—Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

UU-T-111—Tape; Paper, Gummed (Sealing and Securing).

PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Std. No. 187—Identification of Pressed Bends, Forms, Seams and Joints (Sheet Metal).

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-W-12332(ORD)—Welding, Resistant, Spot and Projection for Fabricating Assemblies of Low-Carbon Steel.

Military Standards:

MIL-STD-20—Welding Terms and Definitions.

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the

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procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.1) before production is commenced, a sample of the finished commodity shall be made available to the contracting officer or his representative, either at the contractor's plant or at such other place as designated, for inspection to determine compliance with the provisions of this specification applicable to end production examination and tests. The approval of the sample authorizes commencement of production but does not relieve the contractor of responsibility for compliance with all other applicable provisions of the specification.

3.2 Materials.

3.2.1 *Steel.* The steel used in the fabrication of the costumers shall be of commercial quality, cold rolled, and shall be bright, smooth, free of scale, pits, wrinkles, laps, crimps and buckles affecting strength. Sheets used to form parts shall meet the bend test specified in 4.3.1. All steel tubing shall be seamless or continuously seam welded. The gage numbers specified are the manufacturers' standard gage for steel sheets. The decimal equivalent of the steel thickness mercial tolerances.

is subject to the established standard com-

3.2.2 *Iron castings.* All gray iron castings shall be free of blowholes, porosity, hard spots, shrinkage, cracks, or other defects.

3.2.3 *Zinc.* Die cast zinc-base alloy used for hat and coat hooks shall have approximately the following chemical composition: copper, 0.75 percent; aluminum, 3.75 percent; magnesium, 0.03 percent; iron, 0.10 percent; lead 0.007 percent; cadmium 0.005 percent; tin, 0.005 percent; and sufficient zinc to make 100 percent.

3.2.4 *Bronze.* Bronze used for hat and coat hooks shall have the following chemical composition: copper, 83 percent; tin, 4 percent; lead, 6 percent; and zinc, 7 percent. A tolerance of plus or minus one percent shall be permitted in the composition.

3.2.5 *Finishing materials.*

3.2.5.1 *Primer.* Primer, when used, shall be compatible with the finishing enamel used.

3.2.5.2 *Enamel.* The finish enamel shall be a commercial product of the baking type and the color shall be as specified in 3.2.5.2.1. The enamel shall withstand the test specified in 4.3.2.

3.2.5.2.1 *Color of enamel.* The color of the finish enamel shall be gray, color No. 26134 of Fed. Std. No. 595. Sample panels of the standard color are obtainable, without charge, from the Business Service Center, Federal Supply Service, General Services Administration, Washington, D. C. 20407, or from the Business Service Center of the nearest GSA Regional Office.

3.3 Construction

3.3.1 *Spot welding.* Spot welds shall not be closer to the edge of either adjoining member than 1/2 the greatest dimension of the spot weld, nor shall they be centered closer together than twice the greatest dimension. Squeeze time, weld time and hold time shall be controlled to produce uniform welds that will meet the test specified in 4.3.3. All exposed spot or projection welds shall be filled and ground smooth.

3.3.2 *Definition of terms used herein.*

3.3.2.1 *Welding terms.* All spot welding and electrical resistance welding shall be as defined in MIL-STD-20. Definitions of these terms in MIL-STD-20 are as follows:

Resistance welding: A group of welding processes wherein coalescence is produced by the heat obtained from resistance of the work to the flow of electric current in a circuit of which the work is a part, and by the application of pressure.

Spot welding: A resistance welding process wherein coalescence is produced by the

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heat obtained from resistance to the flow of electric current through the work parts held together under pressure by electrodes. The size and shape of the individually formed welds are limited primarily by the size and contour of the electrodes.
Fusion welding and brazing shall be defined as follows:

Fusion welding: Fusion welding consists of any welding process in which the weld is carried out solely by the melting of the metals to be joined, without any mechanical pressure. The joining is usually accomplished through the fusion of metal deposits from an external source.

Brazing: Brazing is the joining of metals with the aid of a filler metal of a different composition having a melting point above 800 degrees Fahrenheit. Brazing method shall be gasflux, using copper alloy rod, acetylene, and oxygen (or propane) gases.

3.3.2.2 *Fabricating terms.* All bends, forms, seams and joints of the metal specified herein shall be as defined in Fed. Std. No. 187.

3.3.3 *Staff cap.* The staff cap shall be not less than 20 gage (0.0359 inch) sheet steel as specified in 3.2.1, formed into a pyramidal shape and fitted over the top of the staff with the cap sleeve dropping not less than 1/2 inch below the staff height. It shall be permanently fixed thereto by means of welding or blister, one weld or one blister on each side.

3.3.4 *Hooks.* The hooks shall be die cast zinc-base alloy as specified in 3.2.3 or cast bronze as specified in 3.2.4. Casting shall meet the requirements of 3.2.2. The hooks shall be cadmium plated. The design of the hooks shall be as shown in figure 1.

3.3.5 *Base.* The base shall be 2 inches square steel tubing as specified in 3.2.1 with a wall thickness of not less than 18 gage (0.0478 inch.) It shall consist of four individual members fitted over the hub extensions and attached thereto as specified in figure 1. The four points of the base in contact with the floor shall not vary more than 1/32 inch when checked on a commercial surface plate.

3.3.6 *Base hub.* The base hub shall be gray cast iron meeting the requirements of 3.2.2, designed and constructed in accordance with figures 1 and 4, with the four corners of each extension chamfered to allow for the corner radius of the tubing specified in 3.3.5. Alternatively, the hub may consist of component parts of at least 16 gage (0.0598 inch) sheet steel as specified in 3.2.1. The steel hub shall be welded into a single assembly and shall be of the same design and dimensions as the cast hub.

3.3.7 *Base leg cap.* The base leg cap shall be of not less than 20 gage (0.0359 inch) sheet steel as specified in 3.2.1. The cap shall cover the rounded ends of each square tubular base member. It shall be formed to the same radius as the tubing and arc-welded, acetylene- or spot-welded thereto.

3.3.8 *Shoes.* Shoes shall be gray cast iron meeting the requirements of 3.2.2. One shoe shall be fastened to each base member with three No. 8, 5/8 inch long sheet metal screws. The size and shape shall be as specified in figure 3.

3.3.9 *Staff.* The staff shall be 2-inch square steel tubing as specified in 3.2.1 with a wall thickness of at least 18 gage (0.0478 inch) and of sufficient length to permit an overall costumer height of 66 inches. The top of the staff shall be capped as specified in 3.3.1. The staff shall have four coat and four hat hooks as specified in 3.3.2 fastened by 1/2 inch long No. 8 oval head sheet metal screws. The top mounting screws for one set (2 each) of hooks shall be 1-5/8 inches from the top of the staff, not including the cap, and 6-1/8 inches for the other set of hooks. A hub, as specified in 3.3.4, shall be fusion welded or bronzed, (brazed) to the bottom of the staff. A maximum of 1/4 inch out-of-squareness from top of staff to base hub will be allowed.

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3.3.10 *Tolerances.* With exception of base hub, which shall provide for a snug fit of staff and base legs, a plus or minus tolerance of 1/16 inch will be permitted on all component parts. A plus or minus tolerance of 1/8 inch will be permitted in the overall height of the assembled costumer.

3.4 **Pretreatment and finishing.**

3.4.1 *Pretreatment.* All exposed ferrous surfaces shall be cleaned and surface preparation processed in accordance with any of the methods specified in TT-C-490.

3.4.2 *Priming.* Except as otherwise specified herein, all surfaces treated as specified in 3.4.1 shall be coated with primer specified in 3.2.5.1.

3.4.3 *Enameling.* All primed surfaces shall be coated with enamel specified in 3.2.5.2 and baked in accordance with the enamel manufacturer's directions. Total dry film thickness of enamel and primer shall be not less than 1.4 mils. Primer may be omitted from any surface prepared for painting in accordance with type I or type II of TT-C-490 provided the dry enamel film thickness is not less than 1.0 mil. The finish shall level out to produce smooth, uniform exposed surfaces without runs, wrinkles, grits, areas of thin film or no film and separation of color. Finsh panel shall withstand the test specified in 4.3.2.

3.5 **Identification marking.** Unless otherwise specified (see 6.1) each costumer shall be marked in a plain, permanent and inconspicuous manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. When specified (see 6.1) costumers shall be marked in accordance with MIL-STD-130.

3.6 **Workmanship.** The workmanship shall be of a quality to produce a serviceable and neat item that will withstand daily usage. There shall be no sharp exposed surfaces. Welding shall be secure. All exposed welds shall be ground smooth and spatter removed. Corners shall be slightly rounded. The occurrence of defects which affect the appearance or serviceability of the costumer shall not exceed the acceptable quality level specified herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order (see 6.1). The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Preproduction sample inspection.** Examination and testing of the preproduction sample shall be made on the completely fabricated costumer for all provisions of this specification before regular production is started.

4.2.1 *Sampling for inspection and acceptance.* Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1.1 *Component and material inspection.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification.

4.2.2 *End item inspection.*

4.2.2.1 *Visual examination.* Examination of the end item shall be in accordance with the classification of defects specified in table I. The lot size shall be expressed in units of costumers for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit for this examination shall be one completely fabricated costumer.

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TABLE I.—*Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Enamel finish	Wrong color.		X
	Poor adhesion (blistered, checking, or peeling).	X	
	Tacky (not dry to touch).		X
	Runs, sags or foreign matter in finish.	X	
	Not smooth or uniform.		X
	Scratched or bare spots.		X
Construction and workmanship	Any component malformed, dented, buckled or bent affecting serviceability.	X	
	Any component malformed, dented, buckled or bent not affecting serviceability.		X
	Sharp edge or projection that may cause injury.	X	
	Not material specified.		X
	Any component missing.	X	
	Any serviceable component not attached as specified.		X
	Levelness of four contact points of base varies more than 1/32 inch when checked on a commercial surface plate.	X	
	Out-of-squareness of top of staff with base hub more than 1/4 inch.	X	
Welding or brazing	Not fused, burned through, cracked or fractured.	X	
	Not smooth, flash or spatter not removed.		X
	Pitted or not complete.		X
	Incomplete, undercut or missing.	X	
Identification marking	Missing, illegible, incomplete, defective, not permanent.	X	

4.2.2.2 *Dimensional examination.* Inspection will be made of costumers for compliance with dimensions specified herein. Any deviation therefrom shall be classified as a defect.

4.2.2.3 *Inspection levels and AQL's.* The inspection levels and the acceptable quality levels (AQL's) expressed in defects per 100 units shall be as follows:

For examination in—	Inspection level	AQL's	
		Major	Total
Table I	II	2.5	10.0
4.2.2.2 (1 class)	S-4	—	4.0

4.2.3 *Examination of preparation for delivery.* An examination will be made to determine that packaging, packing and marking as required by section 5 of this specification are complied with. Defects will be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-4 and the AQL shall be 4.0 defects per 100 units.

TABLE II.—*Classification of defects*

Examine	Defect
Marking	Incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Any non-conforming component; component missing, damaged, or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of wrapping paper; container flaps, bulging or distortion of containers.

4.3 Tests.

4.3.1 *Bend test for steel sheets.* The steel sheets used for parts to be formed shall withstand a 180 degree bend at room temperature both with and across the grain with-

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out showing any crystallization, cracks or fractures on the outside of the bend.

4.3.2 *Finish test for steel panels.* A panel prepared in accordance with 3.4 shall be bent at room temperature around a 1/4-inch rod to 180 degrees without evidence of cracking, flaking or loss of adhesion of the enamel.

4.3.3 *Spot welds.* Spot welds shall meet the peel test specified in 4.1.4 of MIL-W-12332 (ORD) and button diameter shall conform to table II as specified therein.

4.3.4 *Load test on costumer.* With the costumer in a vertical position, a total load of 400 pounds suspended by means of a suitable harness from the top of the costumer to approximately 12 inches from the floor shall not cause a permanent set of any one of the legs of the base.

4.3.5 *Test on base.* The attached base shall be secured to a vertical plane, and a 60 pound, plus or minus 1 pound, load shall be applied 63 inches from the bottom of the base. A permanent set in excess of 1/4 inch will not be permitted.

4.3.6 *Vertical test for hooks.* The long arm of the hook shall withstand a vertical load of 50 pounds, plus or minus 1 pound, applied 3-1/4 inches plus 1/16 inch from the vertical surface of the staff without producing a permanent set. The short arm of the hook shall withstand a load of 50 pounds, plus or minus 1 pound, applied 1-3/4 inches from the vertical surface of the staff.

4.3.7 *Side deflection test for hooks.* With the staff of the costumer and two hooks in a horizontal position, a load of 25 pounds, plus or minus 1 pound, shall be applied in a manner similar to that specified in 4.3.5 without showing signs of distortion or permanent set.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging and packing.)

5.1 **Packaging.** Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 *Level A.* Each costumer shall be packaged knocked-down to the extent that the base leg members shall be removed. The base assembly hardware shall be packaged in a drawstring type cloth bag which shall be tied to one of the costumer hooks. The base members and the staff with other components assembled thereto shall be completely wrapped in minimum 1/4 inch thick cushioning material fabricated in the form of pads or blankets and may be any one of the following materials.

Macerated paper, plain
Macerated paper, tissue-faced
Shredded paper
Wood wool
Wood fiber felt
Cellulose wadding
Excelsior (sealed ends)
Moulded pulpboard
Laminated cellulose wadding, backed with corrugated fiberboard

Cushioning shall be secured in place with tape conforming to class 2 of UU-T-111 or jute, sisal, manila or cotton twine having a minimum breaking strength of 30 pounds, in such a manner as to prevent any possibility of shifting or becoming detached during transit or storage.

5.1.2 *Level B.* Each costumer shall be packaged as specified in 5.1.1.

5.1.3 *Level C.* Each costumer shall be packaged in accordance with the manufacturer's commercial practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Each costumer, packaged as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style FOL-L of PPP-B-636. Each fiberboard shipping container shall be fitted with a taped one-piece double walled liner having the same bursting strength as the shipping container.

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5.2.2 *Level B.* Each costumer, packaged as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style FOL-L of PPP-B-636. The minimum bursting strength of the fiberboard shall be 200 p.s.i. Each fiberboard shipping container shall be fitted with a taped one piece double-walled liner having the same bursting strength as the shipping container. All flaps of the fiberboard shipping containers shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive.

5.2.3 *Level C.* Costumers, packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to marking required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. 123.

5.3.2 *Military activities.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The metal costumers covered by this specification are intended for general office use to accommodate not more than four hats and four coats.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) If preproduction sample requirement is waived (3.1).

(*c*) If special marking is required (3.5 and 5.3).

(*d*) Inspection record details (4.1).

(*e*) Level of packaging and packing required (5.1 and 5.2).

6.3 Federal specification coverage. Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.4 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Costumers, metallic. (Specify set-up or knocked-down)
Carload minimum weights:
In cars 40'9" or less in length, inside measurements—10,000, 12,000, 15,000, 18,000, 22,000 and 30,000 pounds.
In cars exceeding 40'9" in length, inside measurements — 16,000, 18,000, 21,000 24,000, 30,000 and 40,000 pounds.

Motor:

Costumers, metallic. (Specify set-up or knocked-down)
Truckload minimum weights—10,000, 12,000, 15,000, 18,000, 22,000 and 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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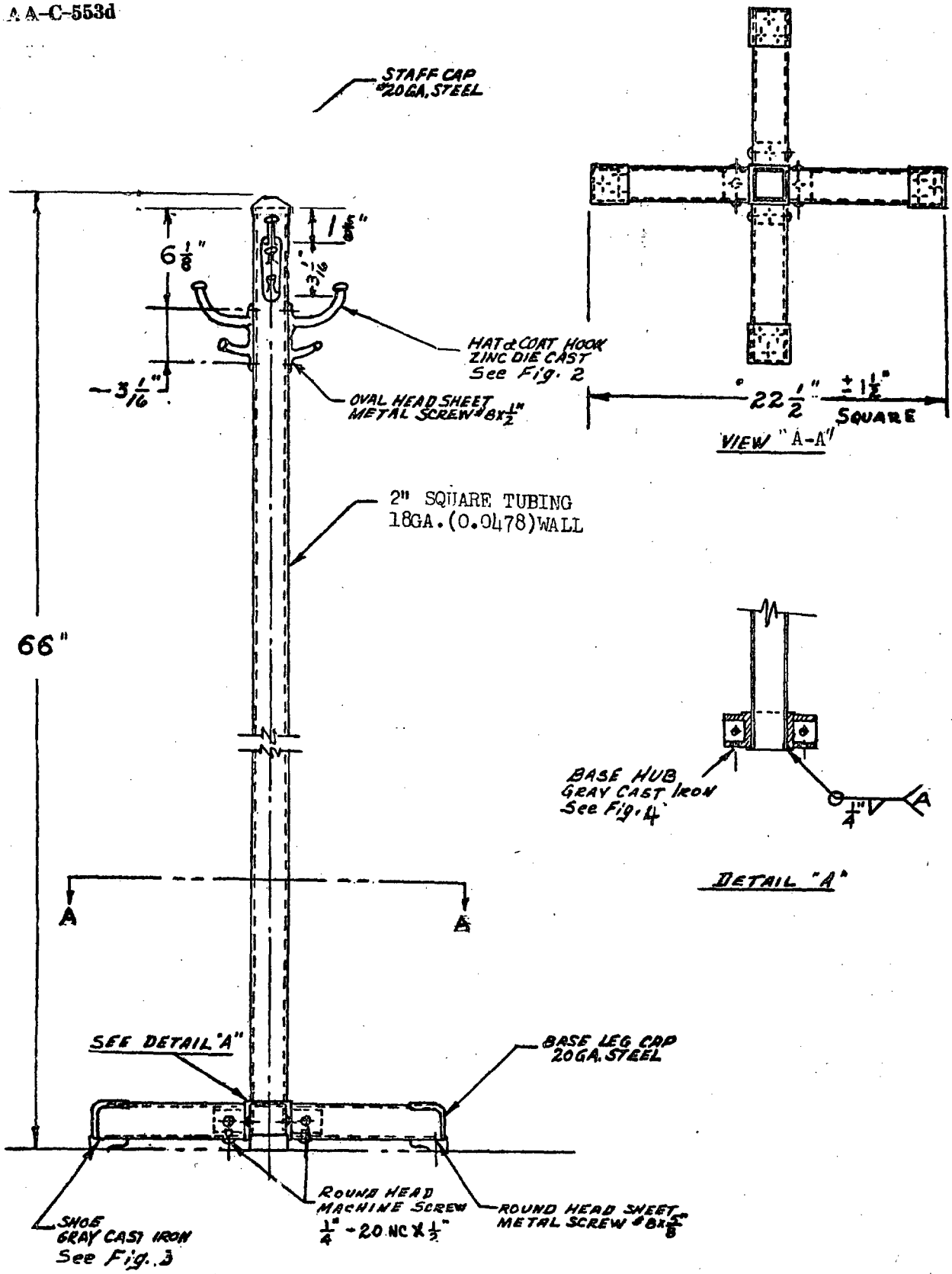


FIGURE 1. Costumer.

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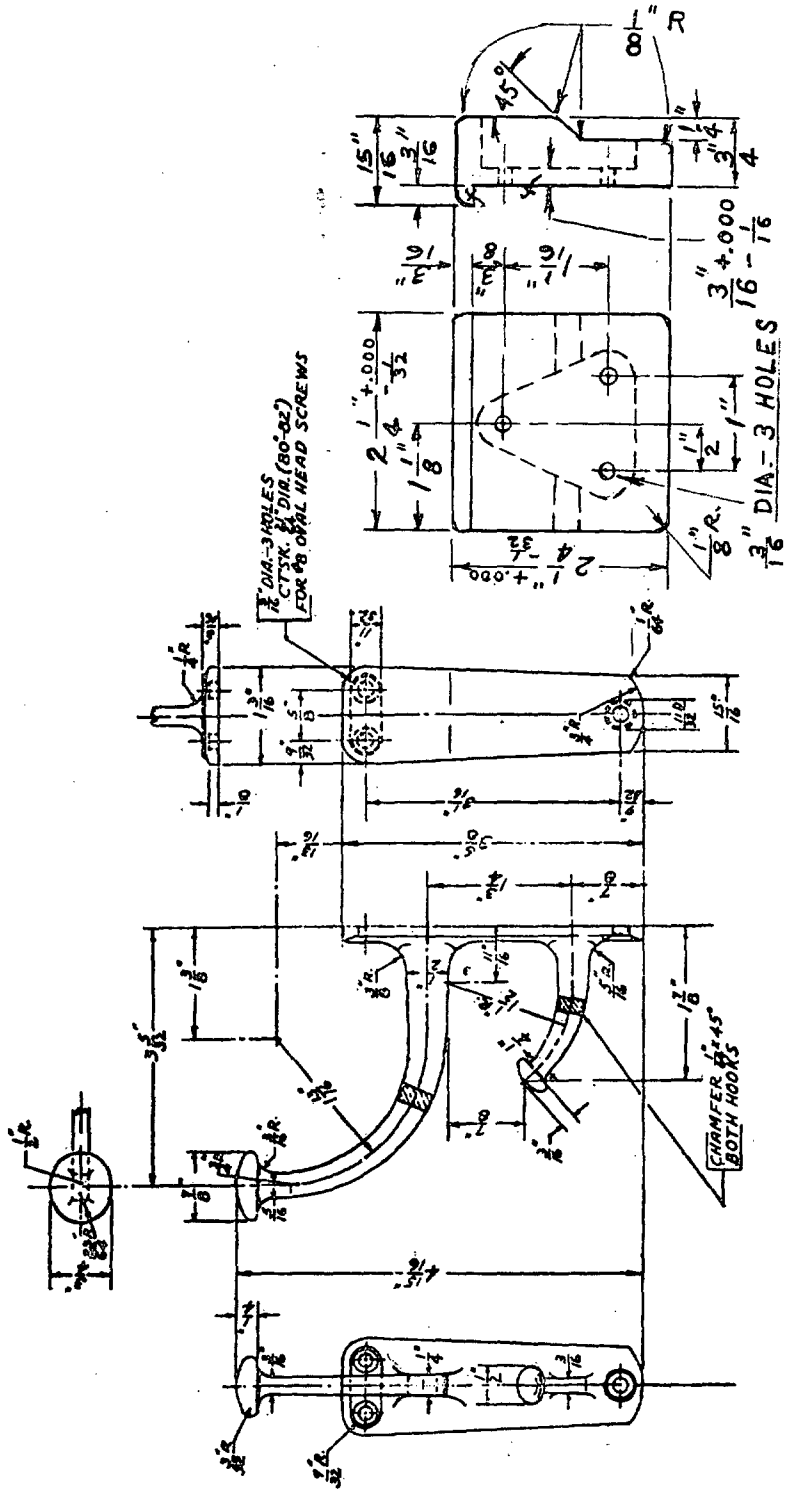


FIGURE 2.

FIGURE 3.

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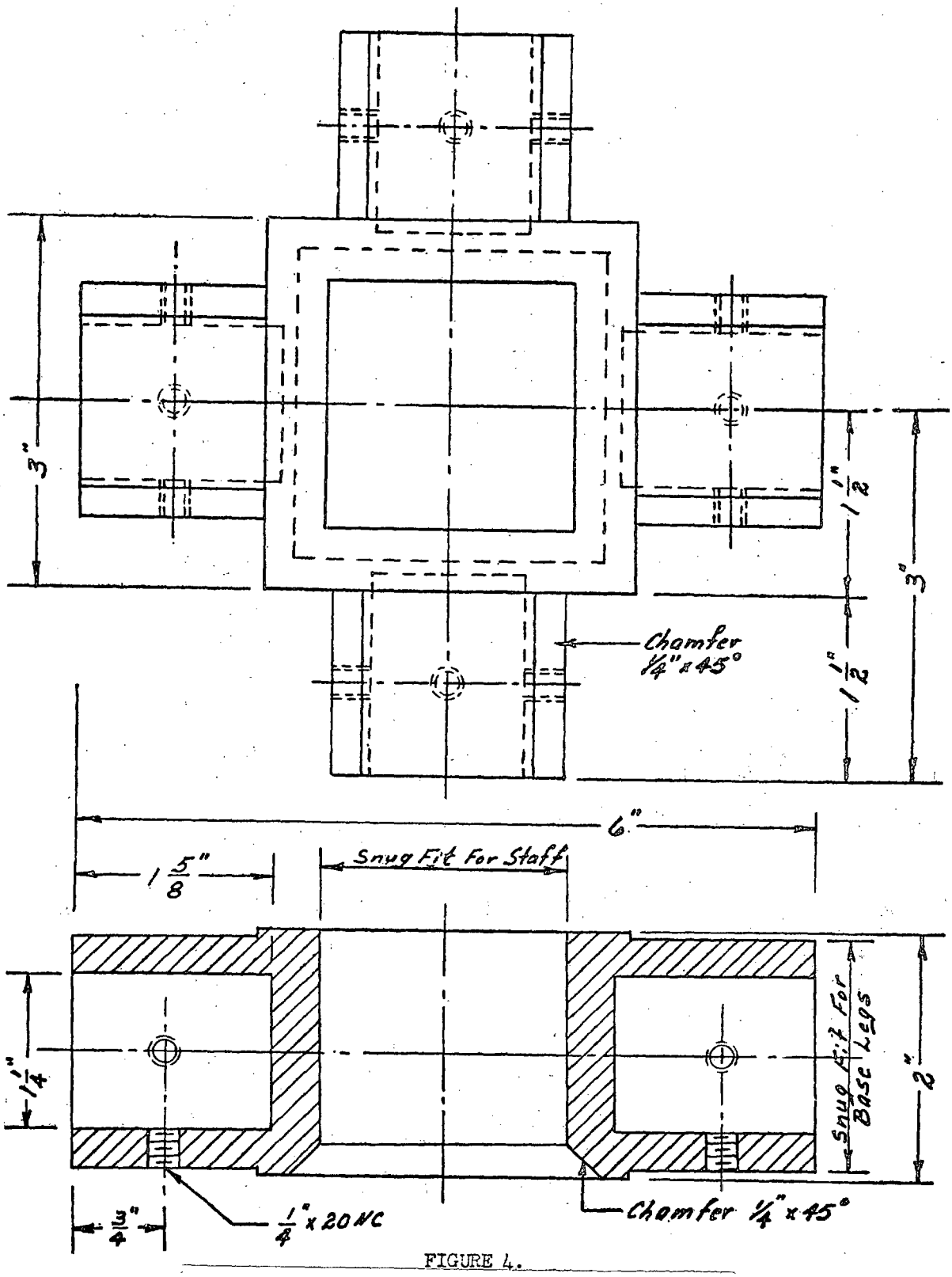


FIGURE 4.

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